

effectiveness and efficiency of machine learning initiatives. Key goals include:

- **Automation:** Implement automated workflows for model training, testing, deployment, and monitoring to reduce manual intervention, minimise errors, and accelerate the development life cycle.
- **Consistency:** Ensure consistency in environments across development, testing, and production stages, reducing the risk of discrepancies that can impact model performance.
- **Scalability:** Design MLOps processes to handle increased workloads and accommodate the growing complexity of machine learning models as organisations scale their ML initiatives.
- **Continuous integration and deployment:** Enable seamless integration of changes to ML models and deploy new versions rapidly, allowing organisations to adapt to evolving data and business requirements.
- **Monitoring and feedback:** Establish robust monitoring mechanisms to track model performance, detect anomalies, and provide timely feedback to data scientists for continuous improvement.
- **Security and compliance:** Implement security measures and compliance checks to address data privacy concerns, ethical considerations, and regulatory requirements, ensuring responsible and legal use of machine learning models.
- **Cross-functional collaboration:** Foster collaboration between data science, engineering, and operations teams to create a cohesive and integrated approach to MLOps, breaking down silos and ensuring a shared understanding of goals and challenges.

Overview of the machine learning life cycle

The machine learning life cycle refers to the end-to-end process of developing, deploying, and maintaining machine learning models. This life cycle typically involves several distinct stages.

- **Problem definition:** Clearly defining the problem or task that the machine learning model aims to solve.
- **Data collection:** Gathering relevant and sufficient data that will be used to train and evaluate the model.
- **Data preprocessing:** Cleaning and preparing the collected data to ensure it is suitable for training and testing.
- **Feature engineering:** Selecting and transforming features in the dataset to increase the model's predictive capabilities.
- **Model training:** Utilising algorithms to train the machine learning model on the prepared dataset.
- **Model evaluation:** Assessing the model's performance using metrics relevant to the specific problem at hand.
- **Model deployment:** Integrating the trained model into a production environment for practical use.

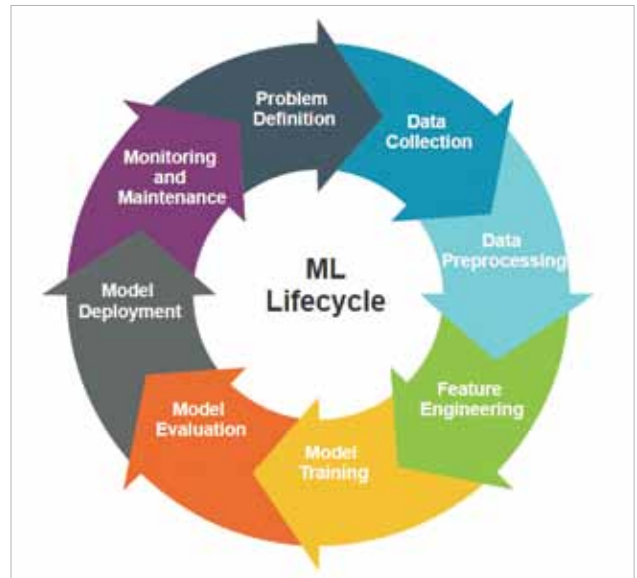


Figure 1: ML life cycle

- **Monitoring and maintenance:** Continuously monitoring the model's performance in the real world and making necessary updates or improvements.

DevOps vs MLOps

While DevOps and MLOps share some common principles, they differ in their focus and application.

	DevOps	MLOps
Focus	Collaboration and communication between software development and IT operations teams	Specifically tailored for machine learning projects, addressing the unique challenges of ML model development, deployment, and maintenance
Goal	Accelerating the software development life cycle, increasing collaboration, and promoting automation	Integrating machine learning workflows seamlessly into the broader DevOps framework, ensuring the reliability and scalability of machine learning systems

Continuous integration and continuous deployment (CI/CD) in MLOps

In MLOps, CI/CD principles are adapted to accommodate the complexities of machine learning workflows, allowing for the seamless integration and deployment of models. This ensures that changes in code, data, or model architecture are tested and deployed efficiently.